

Project Status Report – Week 5

2026-07-06

Project Name: Group 1 Elevator Project

Team Members: Ryan Pratt, Nigel Sinclair, Fergus Page

Project Overall Status: GREEN/YELLOW/RED

Status Overview:

In Phase 1, we developed the elevator project in several domains: the supervisory controller and state machine, the peripheral controllers, CAN communications, web front-end and GUI, external hardware setup, and Linux environment setup. These domains were shared among group members, with individuals leading specific domains and the other group members providing support and input when planning and troubleshooting. We succeeded in creating physical systems and code which combined to yield a functioning elevator system that satisfied the requirements of Phase 1. Our elevator has 3 floor controllers, a car controller, and a supervisory controller – all of which are active and responsive. The supervisory controller runs a state machine coded in a proprietary language, ESE, and the floor/car controllers respond to external and shield button presses and give feedback on shield and external LEDs. The elevator is fully wired, provides state feedback in the terminal, ready to be written into the database, and accounts for communication failures by using a lockout protocol. The supervisory controller implements a heartbeat system to ensure stable function of the control thread and communication thread, and the floor/car controllers implement a similar heartbeat protocol over CAN with the supervisory controller. The GUI is implemented in HTML, CSS, and PHP, and includes a PHP state machine which calculates direction through partially-recursive HTML/PHP form submission. The website includes file interaction, user authentication, and report-logging in PHP, and is ready for database implementation.

Activity Title with start date and also planned completion date and name of responsible team member	Status (Last Week) G/Y/R	Status (This Week)		Plan (Next Week)
		Plan (as planned last week for this week)	Actual Green/Yellow/Red	
Front End GUI/Webpage/Controls/Indicators May 11 – July 6 Fergus	G	Reliable function of numbered floor buttons, directional and level indicators.	G	Implement database interactions. Directional inputs and full elevator controller emulation.
Finite State Machine May 11 – July 6 Ryan	G	Establish possible states, events, and transitions.	G	Database logging and reading. GUI integration.
STM Programming May 11 – July 6 Nigel	G	Floor and car controllers. CAN coms and heartbeat.	G	FIN.

CAN Protocols May 11 – July 6 Nigel & Ryan	G	Develop CAN packaging in both supervisory and auxiliary controller codes. Develop our own protocol. Backward compatibility with class protocol.	G	FIN.
Elevator Hardware Tests May 11 – July 6 All	G	Back-end is working as intended.	G	Testing database functions and response.
RPI/Server/Database Setup May 11 – July 6 Fergus	G	Write Bash scripts to perform RPi environment setup tasks, server setup tasks, and database setup tasks. Write logging handler scripts for each script.	G	Complete database setup script and establish server integration.

Guidelines:

- Overall Status: Green is the best, Red is the worst, and Yellow is in between. You may add + or – suffix to indicate the status more accurately.
- Planned completion date and name of responsible team member: indicate if these are unchanged or changed/new.
- You have to compare your status in three ways:
 1. Status of this week versus the status in last week
 2. Status of this week versus the plan for this week
 3. Planned completion date and responsible person (this week) versus the planned completion date and responsible person (last week)